Data Communication & Networking

Multiplexing of Signals

Mr. K.D.C.G Kapugama

Department of Computer Science

Faculty of Science

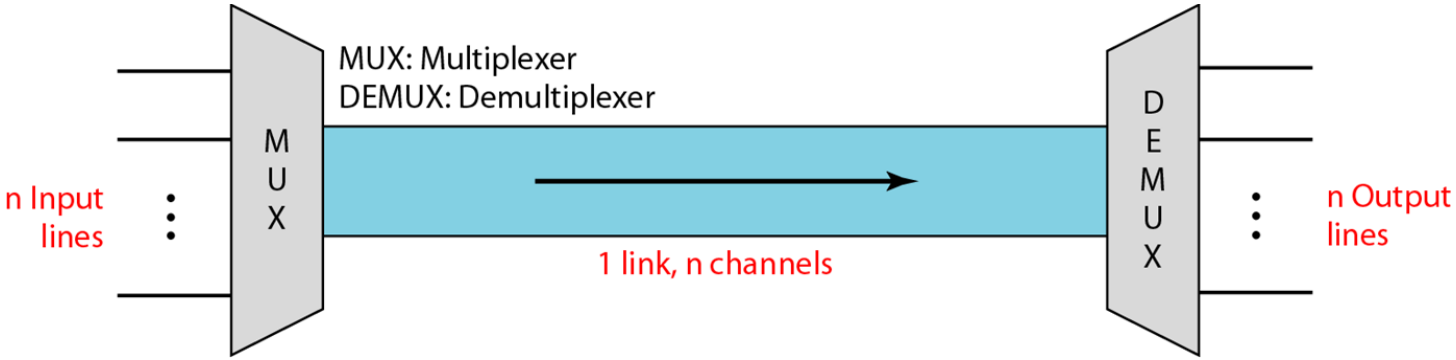
University of Ruhuna

Multiplexing

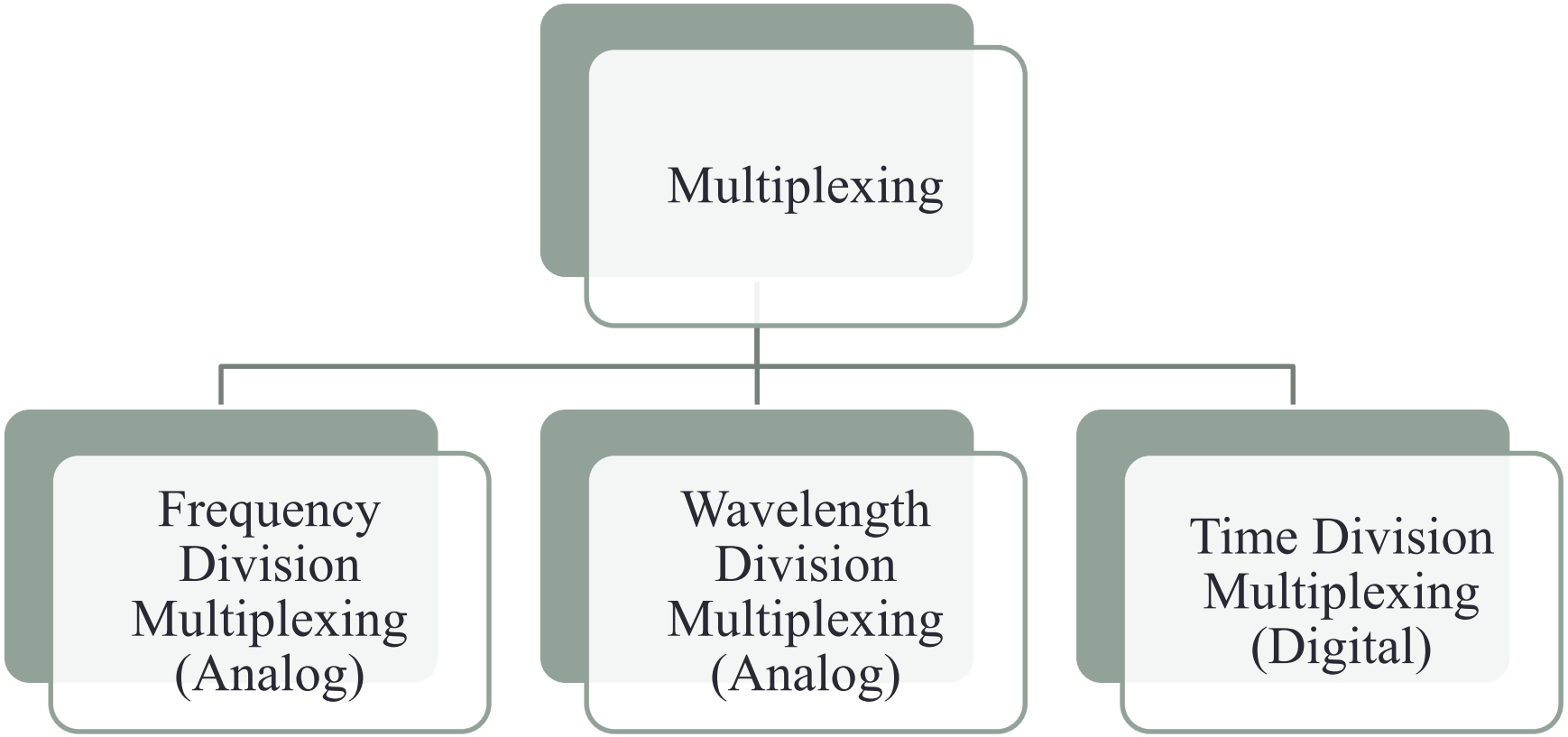
- Most of the individual data-communicating devices typically require modest data rate.
- But, communication media usually have much higher bandwidth.
- So, two communicating stations do not utilize the full capacity of the data link.
- When the bandwidth of a medium is greater than individual signals to be transmitted through the channel, a medium can be shared by more than one channel of signals.
- **Multiplexing** is the process of making the most effective use of the available channel capacity.

Multiplexing

- Multiplexing is the set of techniques that allows the **simultaneous transmission of multiple signals** across a **single data link**.
- Most common use of multiplexing is in long-haul communication using coaxial cable, microwave and optical fiber.



Categories of Multiplexing

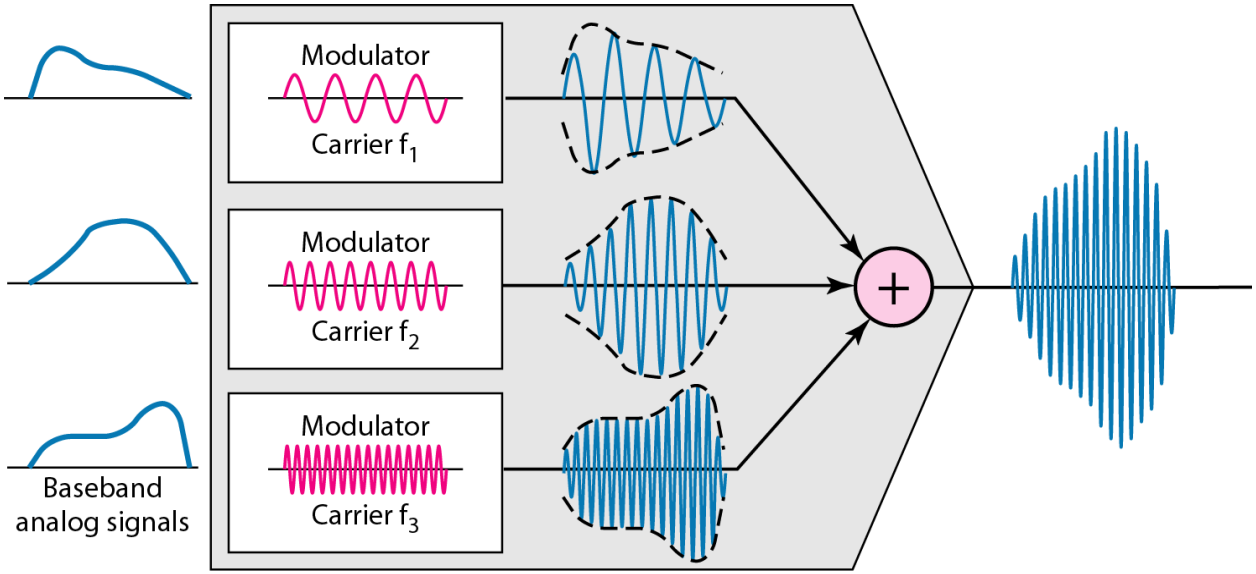


Frequency Division Multiplexing

- The available bandwidth of a single physical medium is subdivided into several **independent frequency** channels.
- Using **modulation**, independent message signals are translated into different frequency bands.
- All modulated signals are combined in a linear summing circuit to form a composite signal for transmission

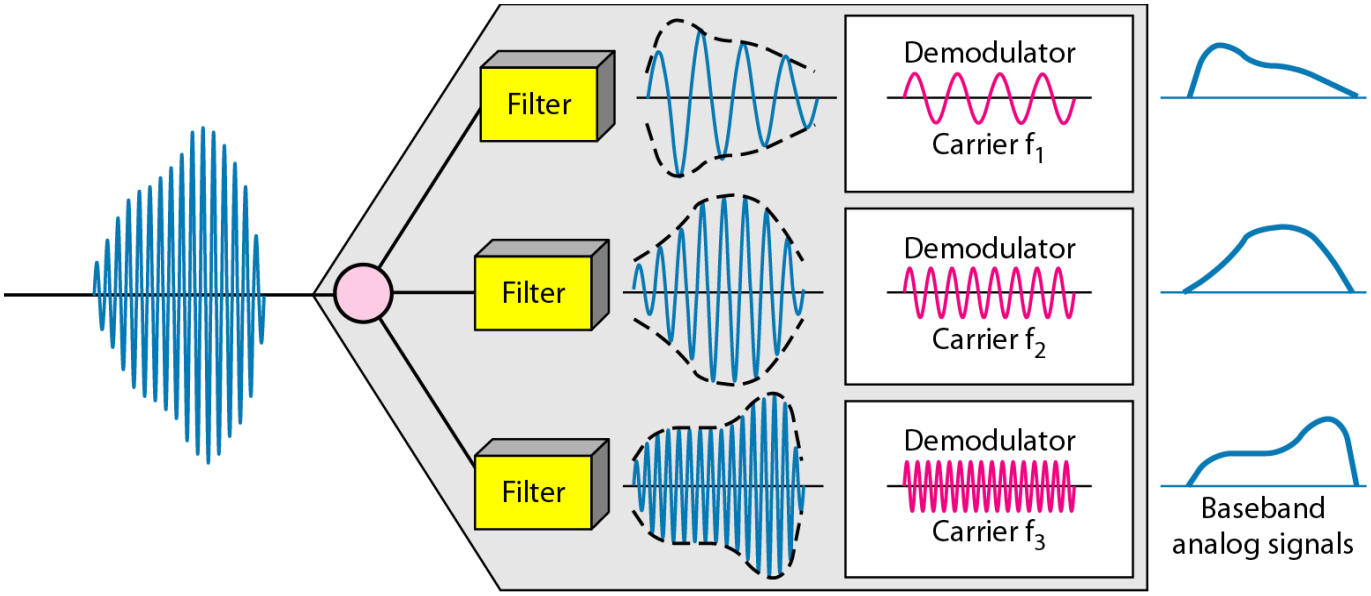
Frequency Division Multiplexing

- **Multiplexing process**
  - Each source generates s signal of a similar frequency range.
  - Inside the multiplexer, these similar signals modulate **different carrier frequencies** (f_1, f_2 and f_3)
  - The resulted modulated signals are combined into signal composite signal.



Frequency Division Multiplexing

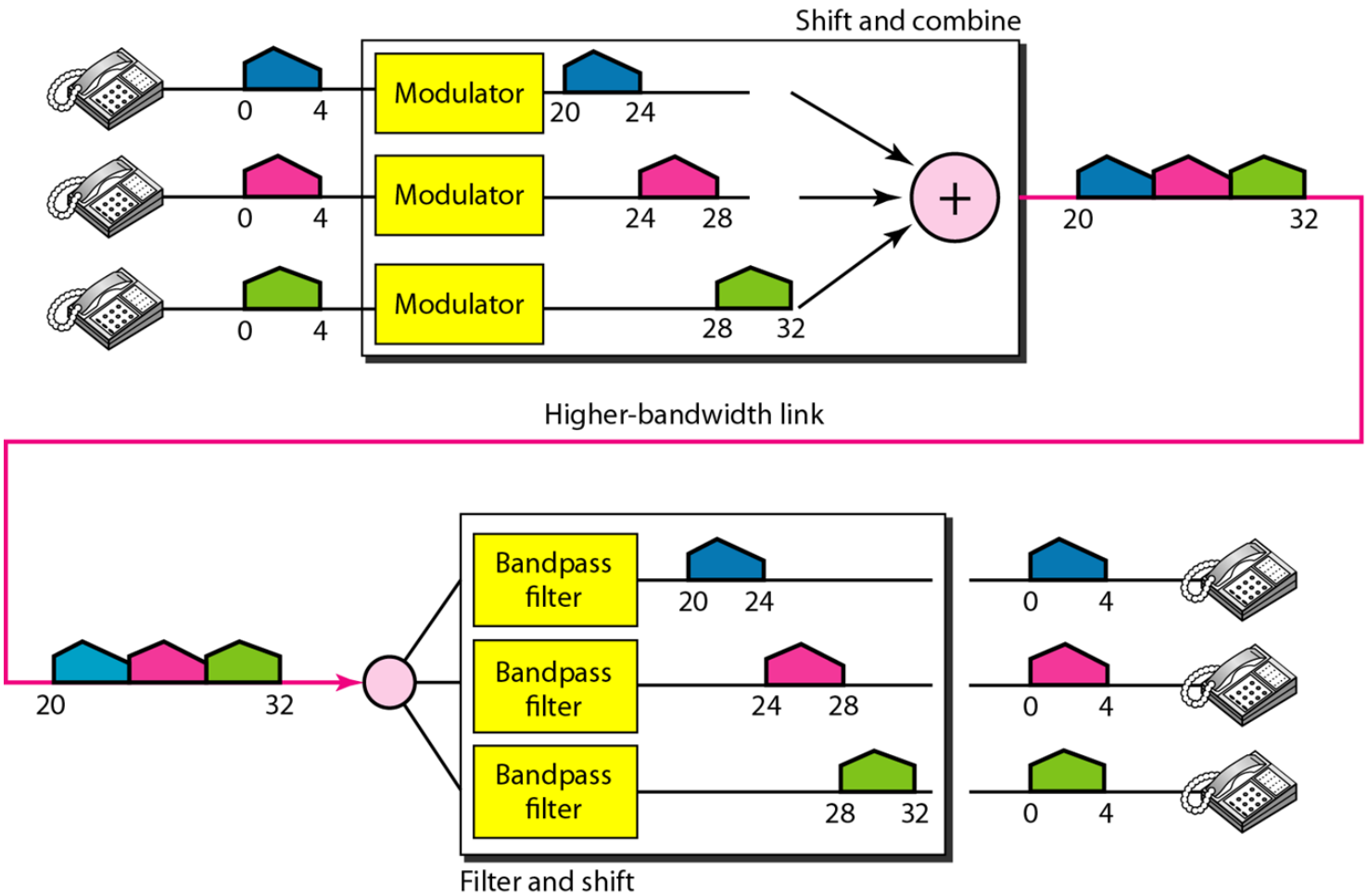
- **Demultiplexing Process**
  - At the receiving end, the signal is applied to a bank of band-pass filters, which separates individual frequency channels
  - Then the individual signals are passed to a demodulator that separates them from their carriers and passes them to the output lines.



Frequency Division Multiplexing

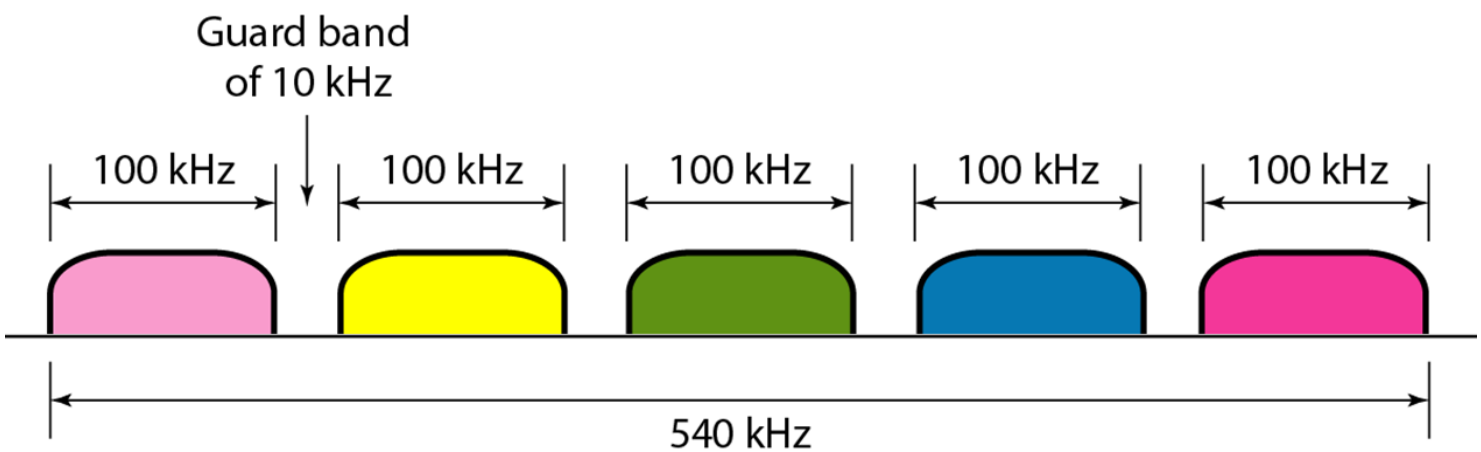
- **Example:**
  - Assume that a voice channel occupies a bandwidth of 4 kHz. We need to combine three voice channels into a link with a bandwidth of 12 kHz, from 20 to 32 kHz. Show the configuration, using the frequency domain. Assume there are no guard bands.
- **Answer:**
  - Range 1: 20 – 24 kHz
  - Range 2: 24 – 28 kHz
  - Range 3: 28 – 32 kHz

Frequency Division Multiplexing



Frequency Division Multiplexing

- Channels can be separated by strips of unused bandwidth – **guard bands** to prevent signals from overlapping.



Frequency Division Multiplexing

- **Exercise**
- Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 6 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth.

- **Answer:**

$$Bandwidth = 4kHz \times 6 + 500Hz \times 5$$
$$= 26.5kHz$$

Frequency Division Multiplexing – Applications

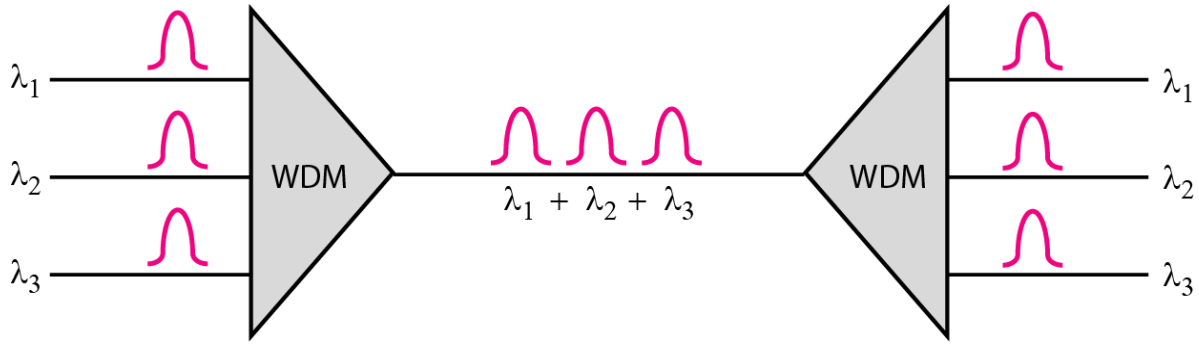
1. AM – FM radio broadcasting. Radio uses air as transmission medium
   - Each AM station needs 10kHz of bandwidth. Each station uses a different carrier frequency, which means it is shifting its signal and multiplexing. The signal that goes to the air is a combination of signals
   - A receiver receives all these signals, but filters (by tuning) only the one which is desired. Without multiplexing, only one AM station could broadcast to the common link, the air.
   - FM has a wider band of 88 to 108MHz because each station requires a bandwidth of 200kHz.
2. Television broadcasting. Each TV has its own bandwidth of 6kHz.

Frequency Division Multiplexing – Applications

3. First Generation of cellular telephones (still operation) also uses FDM
   - Each user is assigned two 30kHz channels, one for sending voice and other for receiving.
   - The voice signal, which has a bandwidth of 3kHz (from 300 to 3300Hz), is modulated by using FM. FM signal has a bandwidth 10 times that of the modulating signal, which means each channel has 30kHz (10×3) of bandwidth. Therefore, each user is given, by the base station, a 60-kHz bandwidth in a range available at the time of the call
   - There is not enough bandwidth in a cell to permanently assign a bandwidth range to every telephone user. When a user hangs up, her or his bandwidth is assigned to another caller

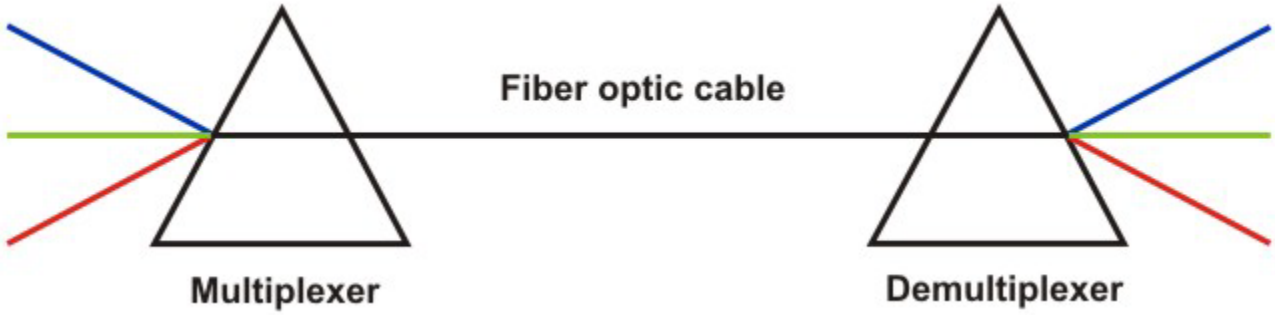
Wavelength – Division Multiplexing

- Designed to utilize the high data rate capability of fiber-optic cable.
- Analog multiplexing technique to combine **optical signals**.
- Very narrow band of light signal from different source are combined to make a wider band of light
- **Optical signals** with different wavelengths are combined, transmitted together, and separated again. (Similar to FDM)
- At the receiver, the signals are separated with the help of a **demultiplexer**.



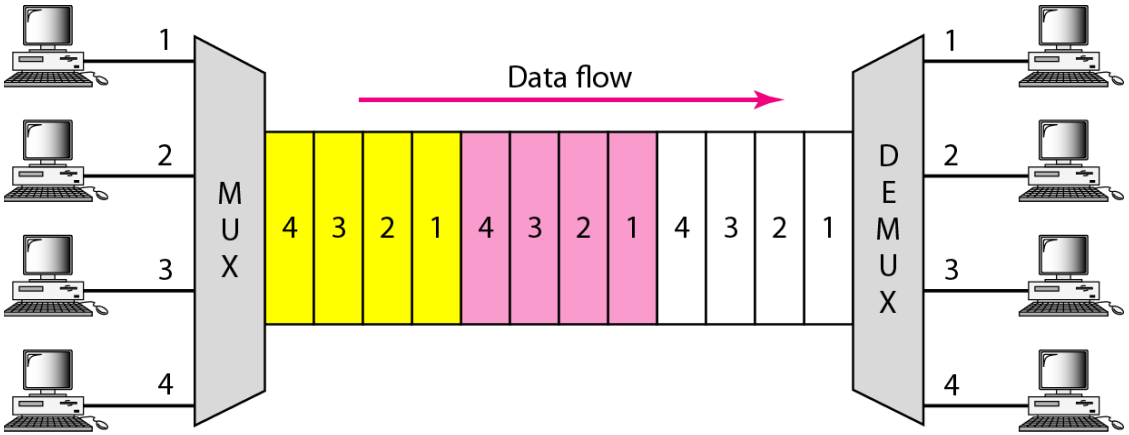
Wavelength – Division Multiplexing

- Multiplexing and demultiplexing of light signals can be done with the help of a **prism**.
- One prism performs the role of a multiplexer by combining lights having different frequencies from different sources.
- The composite signal can be transmitted through an optical fibre cable over long distances, if required. At the other end of the optical fibre cable the composite signal is applied to another prism to do the reverse operation, the function of a demultiplexer



Time Division Multiplexing

- In FDM all signals operate at the same time with different frequencies.
- In Time-division multiplexing, all signals operate **with same frequency at different times**
- Each connection occupies a portion of time in the link
- This is a base band transmission system,



Time Division Multiplexing

- The incoming data from each source are briefly buffered
- The buffers are scanned sequentially to form a composite data stream.
- The scan operation is sufficiently rapid so that each buffer is emptied before more data can arrive
- Composite data rate must be at least equal to the sum of the individual data rates

Time Division Multiplexing

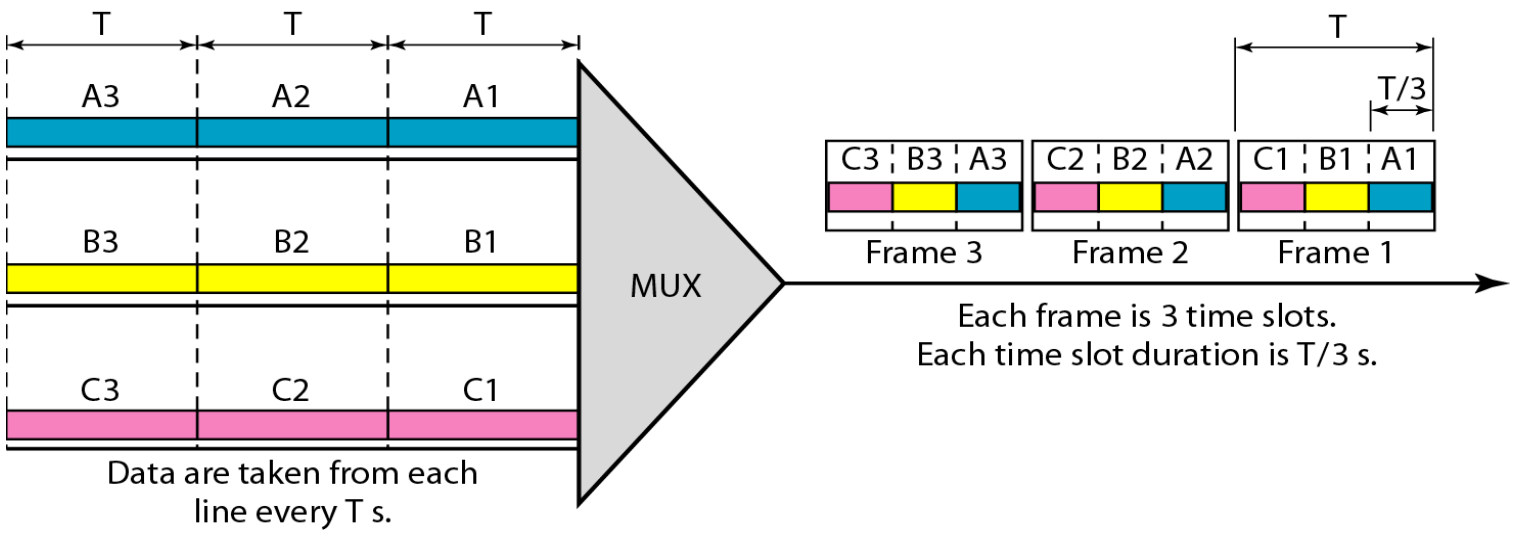
- Two different schemes.
  1. Synchronous
  2. Statistical (Asynchronous)

Synchronous Time Division Multiplexing

- **Time slots and frames**
  - In synchronous TDM the data flow of each input connection is divided into units, where each input occupies one input time slot.
  - A unit can be 1 bit, one character, or one block of data.
  - Each **input unit** becomes one output unit and occupies one **output time slot**.

Synchronous Time Division Multiplexing

- Round of data units from each input connection is collected into a frame



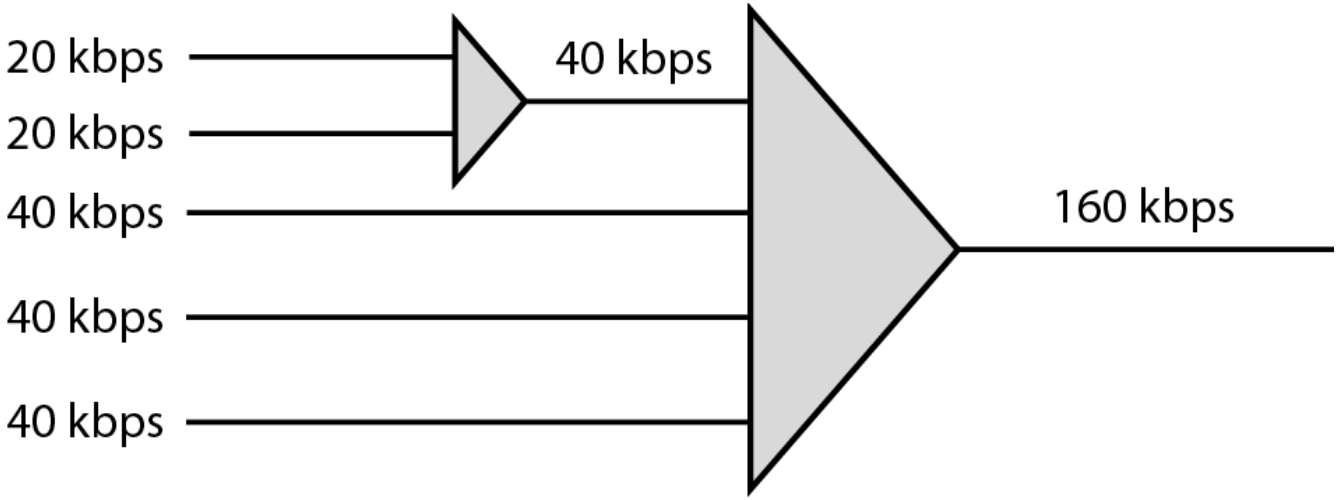
- The data rate of the link is n times faster, and the unit duration is n times shorter.

Synchronous Time Division Multiplexing

- If the data rate of the input channels are not the same, the following three strategies or a combination of those can be used.
  1. Multilevel multiplexing
  2. Multiple slot allocation
  3. Pulse stuffing

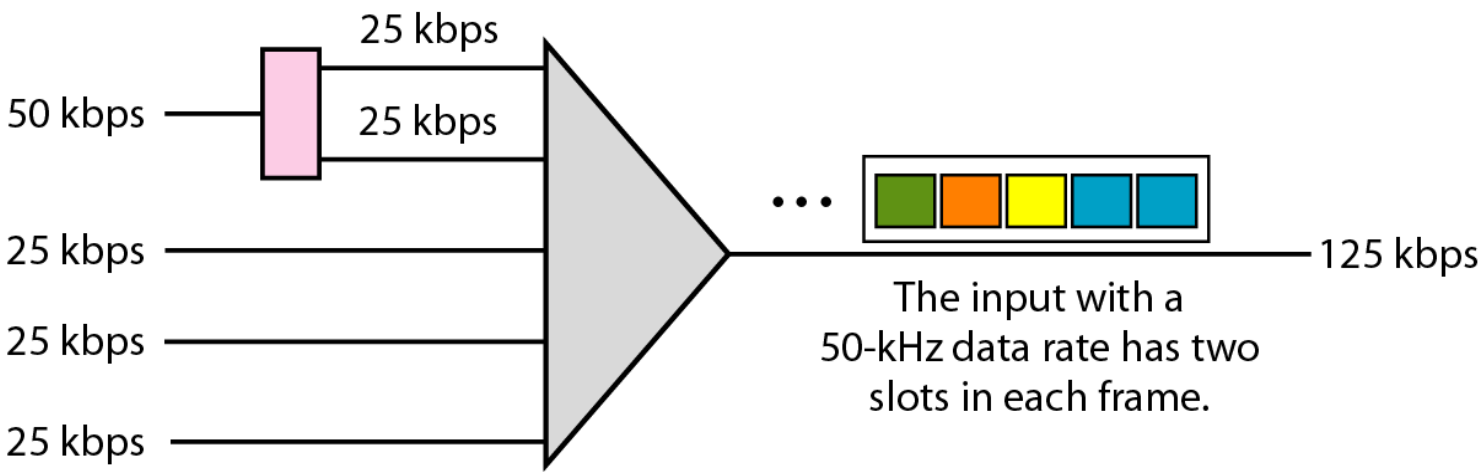
Synchronous Time Division Multiplexing

- **Multi-level Multiplexing**



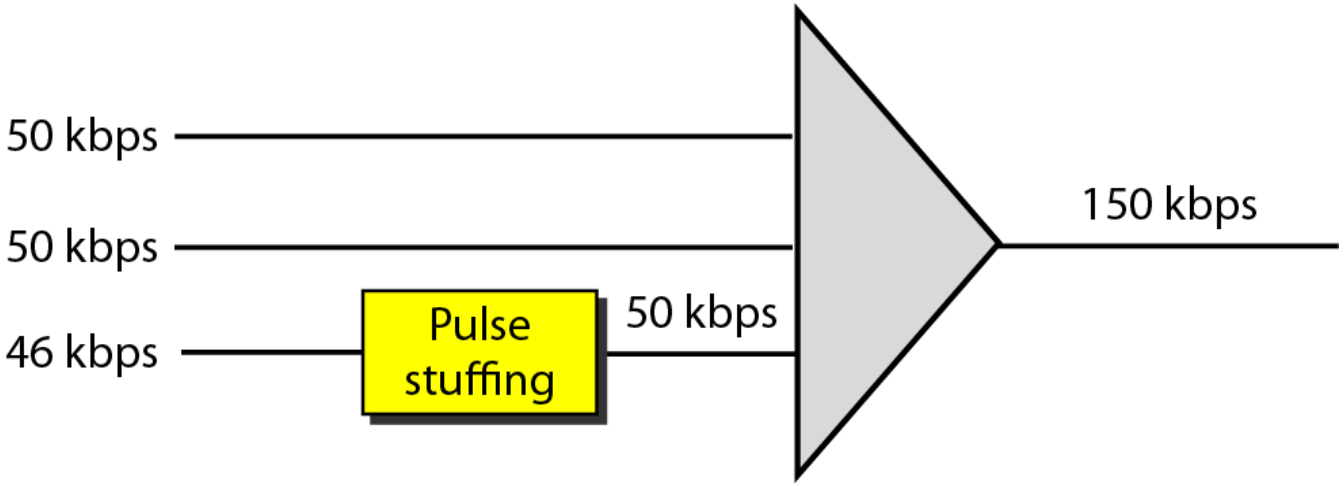
Synchronous Time Division Multiplexing

- **Multiple Slot Allocation**



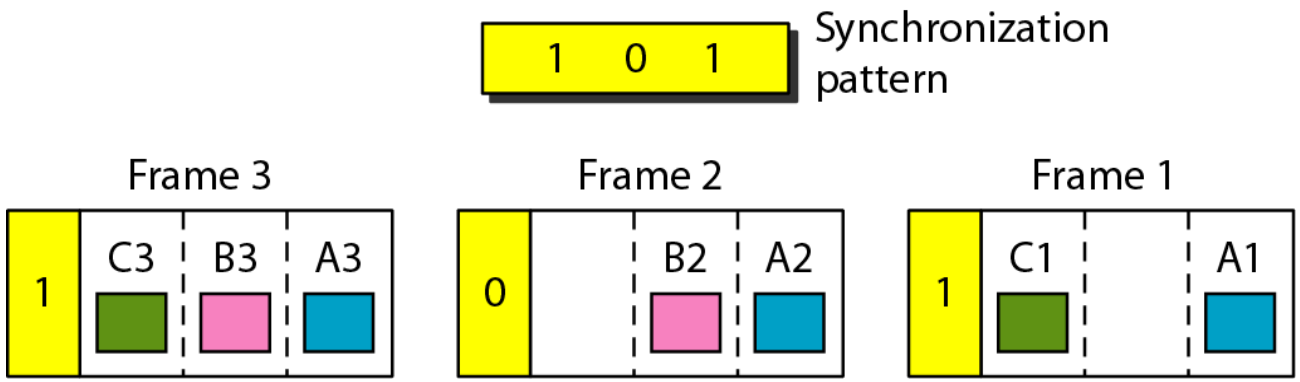
Synchronous Time Division Multiplexing

- **Pulse Stuffing**



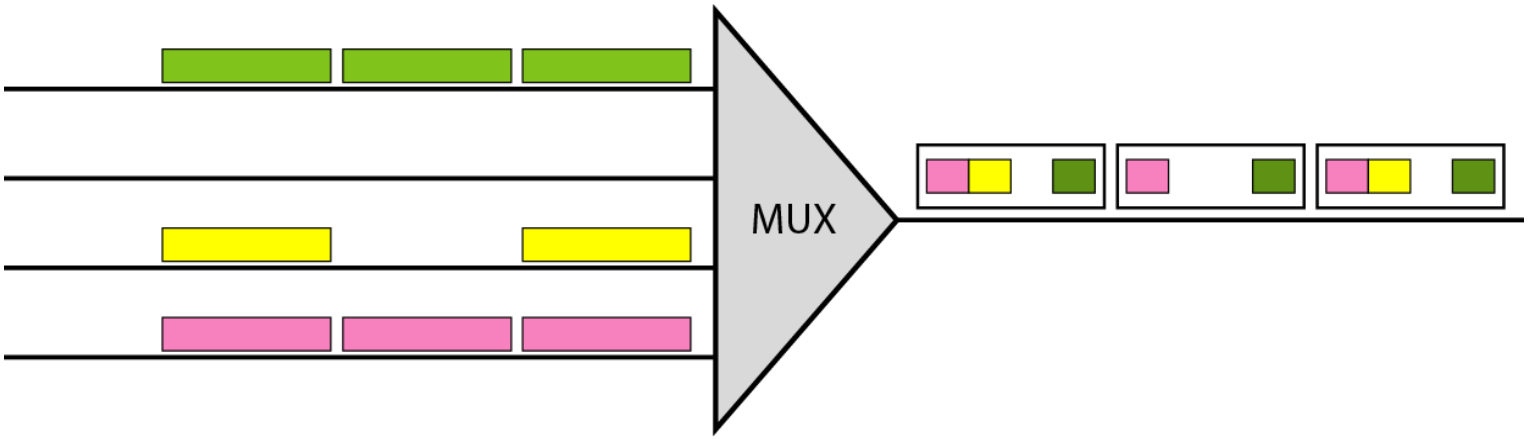
Synchronous Time Division Multiplexing

- One or more synchronization bits, called **framing bits**, are usually added to the beginning of each frame.
- These framing bits help the demultiplexer to separate time slots accurately.



Statistical Time Division Multiplexing

- In synchronous TDM, a slot is reserved for each input in the output frame.
- If a particular input line has no data to transmit, the corresponding time slot in the output frame is wasted. So synchronous TDM can be inefficient.



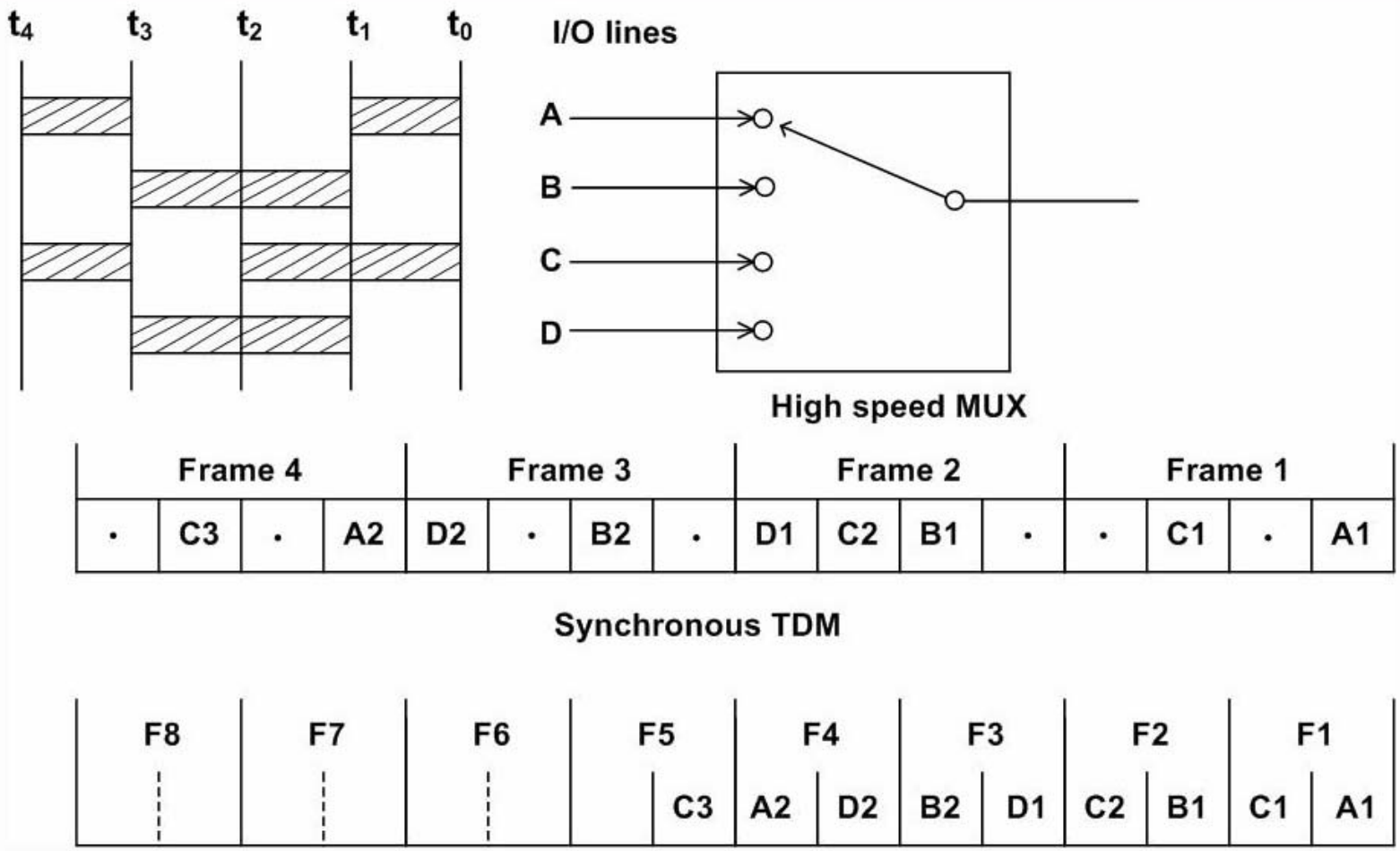
Statistical Time Division Multiplexing

- Statistical Time Division Multiplexing is an efficient alternative to synchronous TDM.
- This is also known as **Asynchronous Time Division Multiplexing.**
- Time slots are **dynamically** allocated on **demand to separate input channels**, thus saving the channel capacity.
- In statistical TDM, the number of slots in each frame is less than number the number of input lines.

Statistical Time Division Multiplexing

- As with synchronous TDM, statistical multiplexers also have many I/O lines with a buffer associated to each of them.
- During the input, multiplexer scans the input buffers, collecting data until the frame is filled and send the frame.
- At the receiving end, the demultiplexer receives the frame and distributes the data to the appropriate buffers.

Statistical Time Division Multiplexing



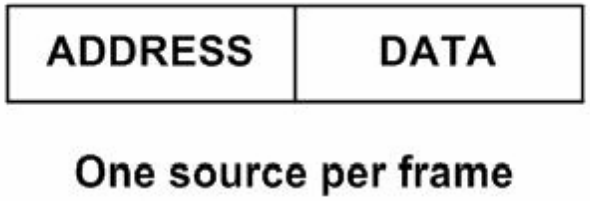
Asynchronous TDM

Statistical Time Division Multiplexing

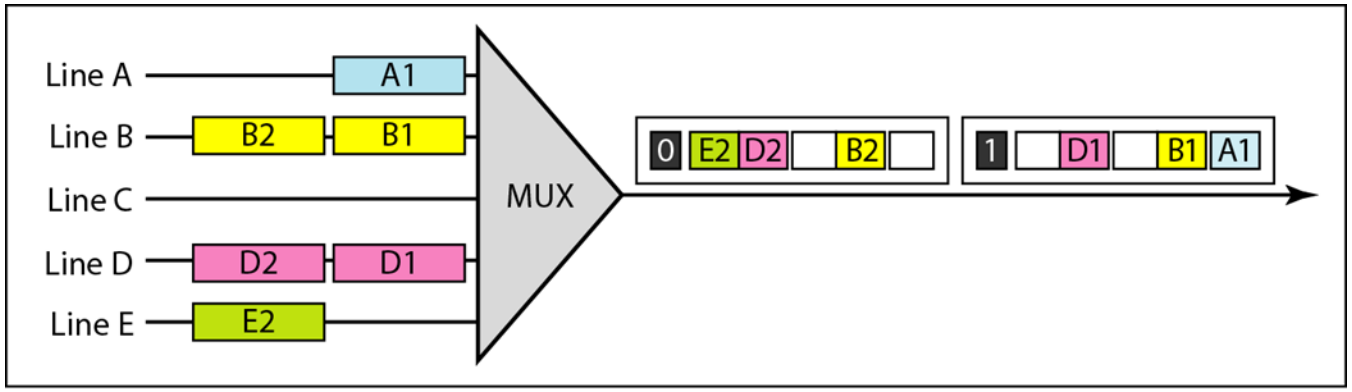
- **Addressing**
  - As there are no pre assigned or reserved slots for input lines, there is no fixed relationship between the inputs and outputs.
  - Thus, it is needed to include the address of the receiver inside each slot to indicate where it is to be delivered.
  - Simply n bits to define N different output lines where

$$n = \log_2 N$$

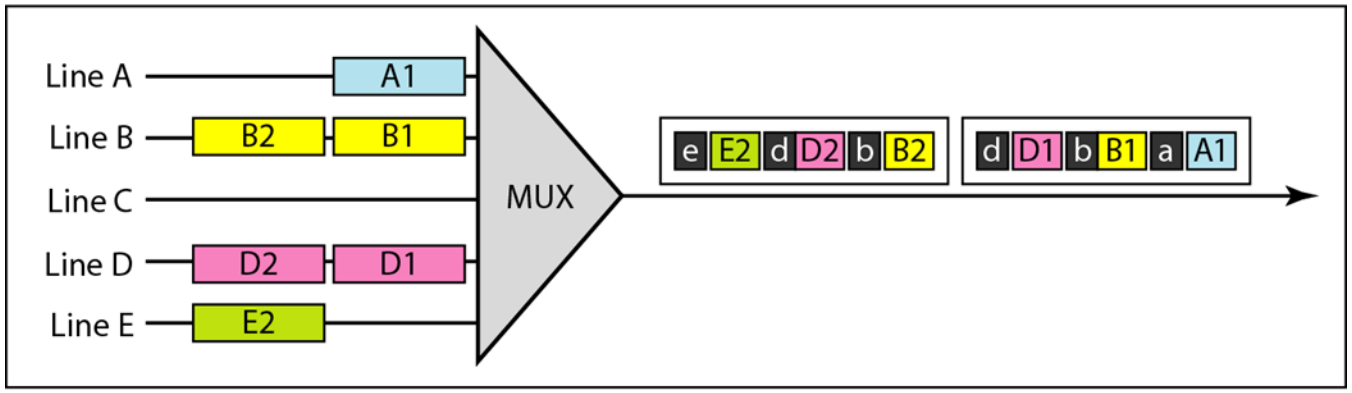
  - For example, for 8 different output lines, 3-bit addresses required.



Statistical Time Division Multiplexing



a. Synchronous TDM



b. Statistical TDM

Statistical Time Division Multiplexing

- **Benefits**
  - Slots are fully utilized leading to smaller time for transmission.
  - Better utilization of bandwidth of the medium
- **Drawbacks**
  - Including address information to the slots leads to more overhead per slot. Relative addressing can be used to reduce the overhead.